GENERAL SPECIFICATION G3.42

Methods of Construction for Sewer Pipes

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G3.42 METHODS OF CONSTRUCTION FOR SEWER PIPES

G3.42.1 Introduction

(a) General

(i) The Contractor shall note that the sewer mains shall be constructed by the methods of open trench excavation or pipe-jacking. Under no circumstance will the Contractor be allowed to change the method of construction specified unless approved by the S.O..

(ii) In all cases the Contractor shall provide for all necessary working shafts, the design of which shall be to the approval of the S.O., and maintain same for the whole duration for which the shafts are left open. Where a working shaft is used for the construction of a chamber, the Contractor shall allow for this in his rates for the construction of the manhole/chamber. Additional working shafts may be sunk with the approval of the S.O. but will not be paid for.

(iii) The Contractor shall allow for any difficulty and inconvenience or site constraints to be encountered during construction and is held to have visited the Site to ascertain the existing condition. The Contractor may propose alternative method of construction if the conditions of the Site is deemed not suitable for pipe-jacking method. The Contractor shall consult and obtain the approval from the S.O. and all relevant authorities before commencement of works.

(iv) Dewatering or lowering of the natural groundwater table shall not be permitted.

(v) Where the term pipe jacking is used in this Section, it refers to laying the pipe by micro tunnelling method. The terms pipe jacking, tunnelling and micro tunnelling is used interchangeably throughout this Section.

(b) Method of Construction – Excavation Generally

(i) The Contractor shall be deemed to have visited and examined the Site carefully and to have ascertained the nature thereof and the kinds of materials to be excavated. Provision must be made in the pricing to cover for all anticipated types of materials to be excavated and the costs of road open applications from all the relevant authorities.

(ii) All excavation shall be carried out to the required lengths, breadths, depths, inclinations and curvatures as may be necessary for the construction of the Works or as shown in the Drawings, in whatever material that may be found. The pricing for excavation shall apply to any type of earth likely to be encountered i.e. soft earth, sand, soft clay, hard clay, tree roots, rock etc. The surplus soil excavation shall be disposed of to the appropriate dumping Sites.

(c) Shoring of Excavation

(i) The term “shoring” shall cover all necessary methods of support including the use of timber, concrete, steel sheet piling, steel trench sheeting, diaphragm wall as specified or such other materials and methods as may be approved by the S.O..

(ii) All “shoring” Works must be designed by the Professional Engineer (Geotechnical, if applicable) engaged by the Contractor and checked by the Accredited Checker (Geotechnical, if applicable) engaged by the Board in accordance with the Building Control (Amendment) Act 2007. The Professional Engineer (Geotechnical, if applicable) must visit the Site on a regular basis to ensure and certify the adequacy of the construction of the Works designed by him. The Professional Engineer (Geotechnical, if applicable) must also monitor the Temporary Works to ensure that the performance is consistent with his design and propose remedial measures when instrumentation monitoring readings exceed “Alert” or “Work Suspension Level”.

(iii) Trench excavation shall not proceed too far ahead of pipe laying especially in metalled road where the trench shall not advance more than 10m ahead of the pipe laying.

(iv) Backfilling and reinstatement works shall not be delayed unnecessarily, and shall not lag the pipe laying by more than 30m. Road reinstatement shall not lag the backfilled trench by more than 50 meters.

(v) For road crossings, each length or cut length of pipe must be laid, jointed, protected and backfilled before proceeding with the excavation for the next section of pipeline for the crossing. The cycle of excavation, pipe laying, jointing, protecting and backfilling for each section of pipeline must be carried out non-stop to completion, until the full crossing has been successfully completed.

(vi) For pipe laying along a busy street especially in the City Area, excavation by machine and laying of pipes or specials by mobile crane shall not be carried out during peak traffic hours but preferably at night and during weekends. Pipe laying shall be carried out in sections not exceeding three pipe lengths at a time. Each section shall be completed and backfilled before proceeding with the next section.

(vii) The Contractor is deemed to have included the above in the pricing of the Tender.

(d) Excavation in Rock

(i) Rock shall be defined as very strong material, in accordance to BS 5930: 1999. The Contractor shall ensure the suitability of machines or shields, which he will provide for the scope of Works in the Contract, for the mixed ground condition, including rock of uniaxial compressive strength more than 150 MPa. The Contractor is deemed to allow for excavating through rock in his rates for pipe-jacking or open-trench method.

(ii) Any unsound areas of rock shall be reported to the S.O. immediately together with a record of the Contractor’s action. The face shall not be advanced until the area is properly secured.

(iii) Any face of exposed ground, where excavation is to be disrupted or discontinued for whatever reason shall be supported by timber, rock-bolts, sprayed concrete or other means agreed with the S.O..

G3.42.2 Excavation for Tunnels

(a) The Contractor shall be responsible for the safety and security of excavations at all times during the execution of the Contract.

(b) The Contractor shall present to the S.O. details of his proposed methods for excavation support and spoil removal. No excavation shall take place until the S.O.'s agreement has been obtained. Such agreement shall not relieve the Contractor of any of his obligations under the Contract.

(c) Excavation shall be carried out in a uniform and controlled manner; over-cutting shall be kept to a minimum consistent with the need to maintain the necessary clearance for construction of the Works.

(d) The invert of the tunnel shall be protected against damage and deterioration which may be caused by construction traffic. Any other surfaces which deteriorate or are damaged shall be made good to a standard agreed with the S.O..

(e) Excavation shall be carried out in sections limited to such lengths, depths and widths as may be safely executed having regards to all the circumstances and as appropriate to the ground conditions and the equipment and method of construction being used.

(f) In water-bearing strata, the Contractor shall use such methods and take such steps as are necessary to control flows and maintain the stability of the excavation.

(g) When necessary to ensure the safety and security of the Works, excavation shall be continuous by day and night.

(h) Weekends and public holidays will require the Works to be made safe and inspected by the Contractor at intervals agreed with the S.O..

(i) Any voids formed during the excavation process by machine overcut slips, falls of materials, overbreak and Temporary Works shall be filled completely with grout, concrete, or other approved durable material, commensurate with the void being filled.

(j) Where the Contract specifies limits to surface settlement and/or protection in respect of existing services or structure, the Contractor shall provide calculations demonstrating that the method of excavation will result in compliance with those requirements. Details of the monitoring arrangements which are proposed for the recording of movements and the verifications of the degree of any settlement or damage to services or structures shall be in accordance with the specifications for instrument and monitoring works.

(k) Where agreed or required by the S.O., temporary support shall be left in the Works. Generally, untreated timber shall not be left permanently in the Works.

(l) Wherever possible, the volume of excavated material shall be measured and recorded as Works proceed.

(m) All excavation shall be carried out to a profile as close as possible to the theoretical excavation line.

(n) The Contractor shall be constantly aware of the possibility of slips and ground movement which may be caused by his method or order of excavation. The Contractor shall at all times maintain on Site material and equipment, for use in ensuring the stability of the face.

(o) The proximity of other tunnels, services and excavations shall be taken into account when determining the method of excavation. The Contractor shall be responsible and included in his rates for ensuring their safety and stability and maintain the same for the whole duration of the Works.

(p) In no condition shall undue loss of ground be allowed at the face, and that there is no run of material from behind supports or lining.

(q) In any exposed ground temporary emergency support shall be available at the face at all times, by means of a full set of cut and marked timber supports kept in good order at all times and stored as close to the face as is practicable and clearly marked as to their function. Alternative methods of support may be proposed by the Contractor.

(r) Where excavation is discontinued for any reason, the Contractor shall take all necessary steps to secure and support the excavation.

G3.42.3 Pipe Jacking

(a) General

(i) The Contractor shall be responsible for inspecting the location and familiarising himself with the conditions under which the work will be performed and with all necessary details as to the orderly execution of the work. The omission of any details shall not relieve the Contractor of full responsibility for the satisfactory installation of the work in its entirety. No monetary or other claims made by the Contractor on the grounds of want of knowledge will be entertained by the Board.

(ii) Where specified in the Drawings or Bills of Quantities, sections of the pipeline shall be laid by pipe-jacking.

(iii) Pipe-jacking is defined as the installation of tunnel lining by jacking of full circle pipes behind a shield or tunnelling machine.

(iv) The method used for pipe jacking works shall be endorsed by the Contractor’s Professional Engineer and submitted to LTA as part of their Road Application Opening Approval. No ‘Heading’ method is allowed. A copy of the method statement is to be submitted to the S.O. for review and approval prior to making an application to LTA.

(v) The Contractor shall be responsible for the design of the pipe jacking as detailed in the Clause “Excavation for Tunnels” of this Specification. All details and calculations shall be submitted to the S.O. for approval.

(vi) Excavation shall be carried out from within a machine or shield equipped with jacks capable of maintaining and adjusting the alignment and shall comply with the Clause “Excavation for Tunnels” of this Specification.

(vii) The contractor shall ensure that the pipe jacking equipment, jacking and receiving shafts, other equipment and ancillaries to be used for the pipe jacking work are able to be installed in the space available on site.

(viii) Equipment and systems shall be designed to provide the forces necessary for the installation of the full pipe string. The Contractor shall be responsible for the design and provision and introduction of intermediate jacking stations.

(ix) The general soil profile along the route of the pipeline is given in the borehole logs, for the Contractor's information only. The S.O. accepts no responsibility for the accuracy, reliability or completeness of this information. It shall be the responsibility of the Contractor to satisfy himself as to the soil conditions along the pipe-jacking route and if he judges it to be necessary, he shall carry out additional soil investigations at his own cost.

(x) In general, the subsoil condition consists of mixed soils, highly permeable and abrasive in nature, and the ground water table will usually be high. The Contractor shall refer to borehole information provided with this tender.

(xi) It shall be the responsibility of the Contractor to maintain stable soil conditions at the jacking face to prevent loss of ground above the jacking operation and movement of the surrounding earth.

(xii) The methods of maintaining face stability and preventing ground movement and subsidence shall be by means of compressed air or other plenum methods where fluid slurry or earth pressure is applied to the tunnel face. Methods which require dewatering of the ground will not be accepted, nor will methods which may lead to significant ground loss. Methods which require workmen to carry out excavation either manually or semi-mechanically in pipes or tunnels smaller than 1200mm in diameter or equivalent will also not be accepted. Any proposed method by the Contractor that requires the use of compressed air must comply with the relevant sections of Workplace Safety and Health Act and the Clauses “Working in Compressed Air” and “Specification for Working in Compressed Air” of this Specification.

(xiii) All tools on the cutter head including the copy and gauge cutters shall be of robust and durable construction in order to minimise the need for replacement during the drive. The Contractor shall expect and allow for variation in soil conditions different from “Borehole Information” provided with this tender and the likelihood of obstruction due to the shallow depth of the sewer pipeline.

(xiv) Where groundwater ingress at the face or elsewhere is sufficient to put at risk the permanent or temporary support, or cause dangerous working conditions, the Contractor shall take action to improve conditions before proceeding. This includes, but is not limited to, relief holes, control of water, recharge wells or ground treatment.

(xv) All excavated material shall be removed from the face of the tunnels immediately after excavation and shall not be allowed to accumulate within the tunnel. Spillages, grout accumulation or dust, oil, waste and materials not essential for the operation, shall be removed from underground.

(xvi) Movement or settlement of structures, utilities and pavement shall be monitored by the Contractor during the jacking operation and reported to the S.O.. If movement or settlement occurs which, in the opinion of the S.O. may cause damage, the Contractor shall take immediate action to prevent further movement, settlement or damage. The Contractor shall repair at his own cost any damage and restore structures or pavements to the satisfaction of the S.O..

(xvii) The Contractor shall engage skilled pipe-jacking operators with a minimum relevant experience of at least 5 years to carry out the pipe-jacking Works in similar ground conditions, particularly in high ground water table and highly permeable soils.

(xviii) The Contractor shall ensure that all key personnel who are responsible for driving, maintenance and controlling the machine have received the necessary training in the duties that they are required to perform. Such training shall include emergency procedures.

(xix) The Contractor shall provide and maintain a complete list of the names of persons, and their duties, responsible for the operation of the machine, who has completed the appropriate training to an accepted standard.

(xx) The Contractor shall deploy a sufficient number of pipe jacking machines, work teams, machineries and equipment, etc. to complete the pipe-lines and manholes within the Contract Period. The Contractor shall demonstrate this with a detailed programme which shall be submitted to the S.O. for approval.

(b) Submission

(i) Plans and details of the equipment, materials and the method of construction to perform and complete the work shall be submitted by the Contractor at the Tender stage and must be approved by the S.O. before commencement of Works. Approval by the S.O. shall not relieve the Contractor of his sole responsibility for the efficacy, reliability and soundness of the method employed in completing the work in a satisfactory manner.

(ii) The Contractor shall submit with his tender, his proposed working method and the names of the manufacturers of his equipment, along with technical catalogues. He shall also submit his past records of successful experience with the proposed method and the names of his jacking crew, along with a summary of their experience in pipe-jacking. The efficacy of the method will be closely monitored during the Contract and the Contractor will be required to adjust or change his method if it fails to meet the Specification.

(iii) The Contractor shall submit method statements for all operations for the agreement of the S.O. before commencement of work. The methods used shall be in accordance with the Guide to best practice for the installation of pipe-jacks and microtunnels published by the Pipe-jacking Association, UK. The statement shall include but not be limited to:

1) the proposed method of transportation and erection of tunnelling machines or shields;

2) the proposed method of commencing the tunnel drives until all the ancillary equipment is installed, including any ground treatment required; temporary thrust arrangements shall be detailed;

3) the method of determining ground conditions ahead of the face and the assessment of the risk of altered ground conditions;

4) the proposed method for monitoring ground loss from the tunnel face during tunnelling, including settlement monitoring;

5) the proposed method of junctioning including dismantling the machines or shields;

6) the method for setting out and controlling the alignment of the tunnels and shafts.

(iv) The Contractor shall also submit a programme for the provision, inspection, testing, transport, erection and commissioning of the TBM or shield.

(v) Before starting work, the Contractor shall submit to the S.O. for approval, a detailed schedule of the entire jacking operation. Approval of such schedule shall not relieve the Contractor of his responsibility to provide a fully satisfactory installation. The Contractor shall include with his submission evidence that he has successfully completed a jacking installation using procedures similar to those proposed.

(vi) The Contractor shall prepare and submit to the S.O. for acceptance, a design and construction procedure that describes clearly the methodology and procedures to be implemented for the installation of the pipes.

(vii) No pipe-jacking work, including shaft excavation, shall commence until the design and construction procedure has been accepted by the S.O. and respective authority.

(viii) The Contractor shall be fully responsible for the design and construction of the jacking and receiving pits, thrusting wall, installation of jacking equipment, sheeting, bracing, etc., and for the efficient execution of the jacking operation. Full details of his proposals, including plant, shield machine, equipment, operating procedures, jacking pit and intermediate jacking stations etc., shall be submitted to the S.O. for acceptance before construction. However, review of the plans shall not relieve the Contractor from his responsibility to provide a safe and satisfactory jacking pit.

(c) Thrust and Reception Pits

(i) Thrust and reception pits and shafts shall be designed and constructed to allow the safe operation of plant, equipment and handling of materials and to withstand all loadings imposed by ground pressure, superimposed loads from surface structures and the maximum anticipated thrust forces. Such calculations including the bearing capacity behind the pits and shafts to receive the thrust forces shall be carried out by a Professional Engineer (Geotechnical, if applicable) for acceptance by the S.O..

(ii) Where Permanent Works accommodate the thrust arrangements, these shall be designed to ensure that the Permanent Work is not damaged.

(iii) In all cases the Contractor shall submit his proposals including calculations to the S.O. for approval. The factor of safety for pipe-jacking shall comply with the manufacturer's recommendations based on the ground conditions and jacking methods.

(iv) The Contractor shall be required to furnish and install and remove to the extent required, thrust blocks or whatever provisions that may be required for backing up the jacks employed in driving the pipe forward. The jacking pit shall also be equipped with steel rails or beams that are securely founded. These rails or beams will be used for placement and alignment of each pipe during the jacking operation.

(v) Special care shall be taken by the Contractor to ensure that the thrust reaction backstop is properly designed and constructed. The backstop shall be normal (square) with the proposed pipe alignment and shall be designed to support the maximum obtainable jacking pressure with a factor of safety of at least two.

(d) Operation

(i) All key personnel shall be experienced in pipe-jacking.

(ii) The Contractor shall be fully responsible for the materials, equipment and facilities required in conjunction with jacking the pipes.

(iii) Before any pipe-jack length commences sufficient pipes and, if required, intermediate jacking station assemblies shall be available to ensure continuous operation.

(iv) The agreement of the S.O. shall be sought for inclusion in the Permanent Works of repaired pipes and standard pipes which have previously been jacked through a drive. Cut pipes shall not be jacked.

(v) The jacking force applied by the thrust pit jacks or an intermediate jacking station shall not exceed the allowable distributed or deflected design load for any pipe being jacked.

(vi) Thrust loads shall be transferred to pipes through a thrust ring which shall be sufficiently rigid to ensure even distribution of the load.

(vii) Intermediate jacking stations shall be inserted no later than when the main thrust pit jacking force reaches a level agreed with the S.O.. This level is typically 66% of the pipe design load appropriate to the amount of joint deflection present in the pipe string, or 66% of the available jacking thrust, whichever is the lesser, unless otherwise agreed with the S.O. during the jacking operation.

(viii) After the jacking operation has begun, the Contractor shall work continuously until the complete length of jacked pipe is installed.

(ix) The Contractor shall provide means to ensure that the pipeline remains stationary when face balance pressure is maintained and when any jacking rams are retracted.

(x) Where chemical stabilization of soil is used, a chemical grout shall be pressure injected into the soil over and ahead of the pipe-jacking to stabilise the soil. The chemical grout used shall have a demonstrated history of success for stabilizing soils similar to that through which the pipe is to be jacked.

(xi) A separation plant shall be provided when using the slurry balance system. The plant shall clean the excavated soil from the water for disposal and return the water back to the cutting face for reuse. The type of separation plant used shall be dependent upon the size of the tunnel being constructed, and the soil type being excavated, and the space available for erecting the plant. In no case shall lagoons be permitted.

(xii) All groundwater encountered during the excavation of the tunnel shall be pressure balanced by the tunnelling machine; no loss of groundwater shall be permitted. The face pressure shall be maintained at all times within ± 20 kPa of the calculated total overburden pressure at the tunnel axis or such other tunnel level as may be agreed with the S.O..

(xiii) It is the responsibility of the Contractor to ensure that the complete tunnels are watertight. If leakage occurs before completion or during the maintenance period, the Contractor shall carry out any remedial work that may be necessary to make the Works watertight all at his own expense.

(e) Lubrication

(i) Lubrication may be applied to the external surface of the pipe to reduce skin friction.

(ii) As part of the operational method statement, a lubricating and ground support fluid, such as bentonite, may be injected into the annulus between the exterior of the pipe and the ground. This fluid shall be maintained under pressure until completion of the drive.

(iii) Where required, the lubricant shall contain an approved additive to limit water loss from the bentonite.

(iv) Where the quantities of bentonite injected exceed the theoretical limits, this shall be reported to the S.O..

(v) Where required under the Contract, grouting of the pipe annulus shall be carried out in accordance with the Clause "Grouting of Pipe Annulus" of this Specification.

(vi) After completion of the jacking operation, the lubricant shall be displaced from between the pipe exterior and surrounding ground by a cement grout.

(vii) The pressure and the volume of grout shall be controlled by the Contractor to avoid pipe damage and displacement of the pipe and soil beyond specified tolerances.

(viii) Grouting shall be accomplished as soon as possible after pipe installation has been completed to prevent any surface settlement due to movement of soil material into the void space or loosened zone around the pipe.

(f) Packing and sealing

(i) Pipe joints shall comply with all relevant provisions of BS EN 1916:2002 ”Concrete pipes and fittings, unreinforced, steel fibre and reinforced”.

(ii) Joint packing material shall be inserted at each pipe joint and at any jacking station in accordance with the pipe manufacturer's recommendations and as specified in the general specifications”.

(iii) There shall be provision to prevent the relative movement between pipes at the joints by the use of steel gaiters or other approved methods during jacking operation. A packing piece of compressible material shall be provided at each joint and shall be securely held before the pipes are lowered into the thrust pit. Details of proposals shall be submitted to the S.O. for approval.

(iv) After completion of jacking, cavities behind jacked pipes shall be filled with grout injected under pressure. All lifting holes and grouting holes shall be sealed with a 1:3 cement: sand mortar with plasticiser, or a purpose made plug.

(g) Monitoring and instrumentation

(i) The jacking system, including any intermediate jacks, shall be capable of continuously monitoring the jacking pressure, the rate of advancement and the distance jacked. Special care shall be taken when setting the pipe guard rails in the pit to ensure correctness of the alignment.

(ii) The Contractor shall maintain site records of jacking loads, line and level measurements, the distance moved and the difference between them. During pipe jacking operation, all main operational data shall be logged onto an industrial standard computer system. As a minimum, the following data shall be continuously recorded:-

1) Total drive length;

2) Deviation from line and level;

3) Roll angle;

4) Steering cylinder stroke;

5) Main jacking station thrust;

6) Cutting wheel torque;

7) Date/time;

8) Provision for operation to add comments to the protocol; and

9) The above data shall be available in a tabular printed record and deviation from the line and level as a graphical printed level.

(iii) The jacking force instrumentation shall be calibrated for each drive by the Contractor. The calibration certificate shall be made available to the S.O..

(iv) Survey control and guidance shall generally comply with sub-Clause on “Survey and Alignment” of this Specification. Further monitoring shall comply with the requirements in specification for Instrumentation and Monitoring works.

(v) The Contractor shall be responsible for monitoring any ground movement associated with the work and making suitable changes in construction methods to control such ground movement and prevent damage or detrimental movement to the work and adjacent structures and pavements.

(vi) Prior to the commencement of any excavation Works, the Contractor shall submit to the S.O. for acceptance calculations of the predicted ground movements associated with the Contractors method of installing the jacked pipes. These calculations shall include all assumptions relating to ground conditions and settlement parameters.

(h) Tolerances

(i) Notwithstanding the specified alignment tolerances, the rate of change of direction in any plane, or combination of planes, shall not exceed the maximum value agreed with the S.O., taking into account the pipe length, diameter, over-cut, jacking loads, and the manufacturer's recommendations.

(ii) The pipes shall be jacked into place true to line and level. The maximum tolerance allowable in the displacement of the centreline of the laid pipe from the design centreline is 50mm in the horizontal plane and 25mm in the vertical plane but there shall be no backfall at any point. Any pipe which is not laid in its correct position must be removed and re-laid or adjusted to obtain its correct position. Changes to line and level shall be gradual. The manufacturer's stated permitted draw or angular deflection on any individual joint shall not be exceeded.

(i) Pipe-Jacking Machine and Shield

(i) The tunnel shall be constructed by pipe-jacking techniques using purpose made Tunnel Boring Machines (TBM) for excavation. The TBM shall have a pressurised closed-face tunnelling system such as a Slurry machine or Earth Pressure Balance (EPB) machine offering controlled support pressure to the tunnel face and allowing for the use of bentonite slurry where required. The TBM must have a provision to deal with ground water in case the access door must be opened in an area where ground water is present.

(ii) The Contractor shall be fully responsible for the selection, design and supply of the machines, shields and ancillary equipment. The complete system shall be purpose designed for the project while being compatible with the anticipated soil and geotechnical conditions. The system shall include TBM, slurry circuit or EPB system, pressure bulkhead, main jacking station, inter-jack stations and automatic bentonite injection system for lubrication of the pipe annulus.

(iii) All TBMs shall be robust with adequate safety margins for the anticipated duty, designed and manufactured to comply with all relevant safety standards. The TBM shall have the minimum requirements as listed in the Technical Schedule.

(iv) The cutter face shall at all times be able to support the full excavated area without the use of ground stabilisation and have the capability of measuring the earth pressure at the face. For slurry shield, balancing of earth and groundwater pressure shall be achieved by use of slurry. A slurry bypass method shall be included to allow for change in direction of flow to be made or isolated. As for EPB shield, balancing of earth and groundwater pressure shall be by the excavated ground in the excavation chamber, with use of ground conditioning agents where required. Both systems shall be capable of incremental adjustment to maintain face stability for soil conditions encountered.

(v) The pressure control system shall automatically maintain the required pressure on the face at all periods when the machine is advancing, and when standing. Control shall be such that the pressure can be adjusted to suit changing face conditions and can be maintained at all times within ±10% of the required pressure.

(vi) Pressure sensors capable of working in air, liquid, spoil/liquid and spoil media shall be mounted on the pressure bulkhead or slurry pipework with pressure gauges in the control cabin reading the pressure in the chamber.

(vii) If machines of diameter 1200mm and above is used, access is required from inside the machine to the front of the shield by means of an access door to replace disc cutters which are worn by the abrasive ground during pipe-jacking. The door can also be used to deal with unforeseen obstructions which may block the advance of the TBM. The likelihood of such obstructions is high due to the shallow pipeline. All tools on the cutter head including the copy and gauge cutters shall be of robust and durable construction to minimise the need for replacement during the drive. The TBM design shall enable replacement of cutter tools to be carried out at the back of the cutterhead by bolting system. The TBM must have a provision to deal with ground water in case the access door must be opened in an area where ground water is present.

(viii) The TBM shall be equipped with an airlock system. Where the machine uses a low pressure compressed air system to gain access to the pressure chamber, this shall comply with the provisions of EN 12110. Work in compressed air shall comply with the Clauses “Working in Compressed Air” and “Specification for Working in Compressed Air” of this Specification.

(ix) The external diameter of the TBM or shield shall be designed to produce minimum overbreak and the least necessary clearance for the proper construction of the Works. The maximum annular space shall not exceed 25mm unless otherwise specified. Design shall consider the horizontal and vertical alignment to be negotiated. Provision shall be made to limit and correct roll of the machine.

(x) All operating functions of TBM or shield, including rate of advancement, shall be in the control or direction of the TBM or shield driver. The TBM shall be capable of controlling shield rotation by means of bi-directional drive on the cutter head. The TBM shall be mechanically articulated to enable remotely controlled steer of the shield.

(xi) There shall be adequate access to all areas of the TBM where work, including maintenance, may be carried out. A clear means of egress in an emergency shall be provided and maintained at all times. Minimum dimensions shall conform to the appropriate EN standard.

(xii) The machine and TBM shall be guided by a laser guidance system, which will allow curve drives, that is acceptable to the S.O. to ensure that the specific curve alignment is achieved during advance. A guidance system, with the ability to predict ahead of the face, installed on pipe-jacking machines with a display to show the position and attitude of the machine is recommended. Line and grade shall be controlled by the guidance system that relates the actual position of the TBM design reference. The display shall be positioned to enable the machine driver to have a clear view at all times.

(xiii) TBMs shall be furnished with a means of controlling orientation. Steering information shall be monitored and transmitted and saved to the operator console. The TBM display shall continuously show and automatically record the position of the shield relative to the design reference, roll, inclination, altitude, rate of advance, installed pipe length, thrust force, cutter head torque, slurry pressure, slurry flow and slurry valve positions. Lasers or other means are to be provided to determine the line, level and square of the shield. Detailed guidance information shall be checked against the tunnel alignment control at regular intervals as agreed with the S.O..

(xiv) The TBM shall have suitable features to ensure that the shield shall maintain line and level and shall be able to make the required adjustments to recover the line and level if necessary. Such features may include an articulated cutter head, programmable copy cutters etc.

(xv) The TBM shall be capable of bi-directional drive on the cutter head wheel to minimise drift and rotation or roll of the pipe during pipe-jacking operation.

(xvi) On completion of the pipe-jack the Contractor shall demonstrate that each straight run of casing pipe shall show a 97 percent full circle of light through the casing when viewed from the adjoining end of line.

(xvii) The Contractor shall demonstrate that the equipment and installation methods shall not damage the pipes or their lining.

(xviii) A lubrication system that injects an accepted lubricant to lower the friction developed on the sides of the pipe during jacking shall be provided.

(xix) The Contractor shall provide for each TBM, spare parts that are considered critical and long lead items. Storage of required spare parts shall be within Singapore and accessible to be delivered to the site within 24 hours. The Contractor shall operate and maintain the machine to the satisfaction of the S.O. and shall maintain an inventory, recommended by the TBM manufacturer, to ensure continued availability of all essential systems.

(j) Probing and Drilling Ahead

(i) Provision shall be made for probing and drilling ahead of the tunnel face to confirm the ground quality and to identify the presence of any water, fissures/strata, discontinuities or voids, services, obstructions, etc. in the ground.

(ii) A sufficient pattern of probe holes is to be provided to minimise the risk of encountering unidentified services, obstructions, water bearing fissures and strata. Probing ahead for up to 40 m should be allowed for, with a continuous overlap of not less than 10m between probe holes ahead of the face. The ability to control any inflow of ground water encountered to a manageable quantity ahead of the face; and stabilise/seal ground to mitigate surface settlement and minimise any delay to the tunnel progress is considered to be of the utmost importance and is to be included in these provisions. An item for this work has been allowed for in the Bills of Quantities.

(k) Spoil Removal

(i) The excavation and disposal arrangements shall be capable of dealing with the full range of materials expected. Generally, the disposal system shall accommodate material produced by the cutting equipment. A trap may be provided for the retention of pieces of material which would otherwise cause damage to or stop up the disposal system.

(ii) Systems incorporated into slurry machines shall be provided with means of accurately controlling and adjusting the density and viscosity of the medium supplied to the pressure chamber and introducing additives where required. Pipework, pumps and separation plant shall be designed to accommodate the maximum rate of advance at which the machine will be progressed. The separation plant shall be such that an assessment of the nature and volume of excavated material can be made.

(iii) Conveyors for disposal systems shall be equipped with emergency trip wires along their entire length. These trip wires should be installed on both sides of the conveyor if access is possible to both sides. All conveyor belts shall be fitted with audible startup warning devices an initial startup delay of 6 seconds. The belts shall be interlocked to prevent partial startup and the possibility of secondary belt overload.

(l) Fire Protection

(i) A high standard of housekeeping shall be maintained within the machine and trailing back up system to prevent the build-up of flammable materials.

(ii) The driver's cabin of remote control machines shall be equipped with fire extinguishers. Fire extinguishers shall also be sited at suitable locations and inspected on a weekly basis.

(iii) The TBM and trailing backup system shall have an automatic fire detection and suppression system which will initiate both an audible and visual warning clearly identified on all parts of the machine and trailing backup system. Manual fire alarm actuators shall also be provided at strategic locations.

(iv) The fire alarm system and static firefighting equipment shall be inspected, serviced and dry tested monthly. A wet test shall be carried out at a frequency of not greater than three months. A detailed record of system status shall be kept.

(v) The Contractor shall carry out a detailed risk assessment of possible fire hazards during the specification stage of the overall fire detection and control system. The Contractor shall seek advice from experienced equipment supplier and shall take due cognisance of any recent developments in component design.

(vi) The Contractor shall ensure that all tunnelling crews are trained in the use of the firefighting system.

(vii) The Contractor shall submit a comprehensive plan concerning Fire Precautions and Evacuations in Tunnels, both during tunnelling and the equipment installation phase, to the S.O. and Fire Safety & Shelter Department (FSSD) of SCDF for acceptance. The following shall be addressed:

1) Burning and Welding;

2) Materials to be used;

3) Fire Fighting Equipment;

4) Fire Fighting Procedures (including fire drill); and

5) Any other activities which may pose a fire risk;

(viii) The contractor is required to comply fully with the latest edition of the Workplace Safety and Health Act.

(m) Equipment Guards and Emergency Stop Facilities

(i) All moving equipment installed shall be adequately guarded to ensure safety against personnel injury and equipment damage. Moving equipment shall be identified with a high visibility paint scheme.

(ii) All strategic moving equipment (e.g. conveyor) shall be fitted with emergency stop buttons and/or emergency stop pull wires. Smaller items of equipment that form part of a larger system (e.g. hydraulic cylinders) shall be protected by power source emergency stops.

(iii) A detailed risk assessment of guarding, warning devices and emergency stop control switches shall be carried out at the time of TBM and trailing backup system design and manufacture. The assessment shall be reviewed during system commissioning underground and shall be subject to regular follow up inspections during normal tunnelling operations. A record shall be kept of all findings and action taken. Those records shall be made available for access by the S.O. at any reasonable time.

(n) Inspection and Testing

(i) Where a new, modified or purpose built TBM is proposed the S.O. shall be permitted to visit the premises of any manufacture for the purposes of inspection.

(ii) The Contractor shall be responsible for the quality of materials used or present within the TBM and must ensure that all materials used or present are adequate for the task they are to perform.

(iii) The S.O. shall be permitted at any stage during manufacture to inspect, examine and test on the manufacturer's premises the materials, workmanship and performance of all plant and components to be supplied under the Contract. Such inspections, examination or testing shall not release the Contractor from any obligation as specified under the Contract.

(iv) When tests are to be made at the premises of the manufacturer or elsewhere, except where otherwise specified, the Contractor shall provide such assistance, labour, materials, electricity, fuel, stores, apparatus and instruments as may be requisite and as may be demanded to carry out such tests efficiently.

(v) New and reconditioned machines shall be assembled at the manufacturer's Works on completion of fabrication or modification and tested to demonstrate that all components operate correctly. Test running will also be required at Site following assembly, prior to commencement of tunnel driving.

(vi) A copy of the manufacturer’s operating and maintenance instruction written in English shall be provided to the S.O. within fourteen (14) days of completion of the TBM manufacturer’s factory testing and commissioning. The machine operators and the Contractor’s chief mechanics, if any shall be trained by the machine manufacture prior to the start-up of the machine.

(o) Survey and Alignment

(i) The Contractor shall set out the Works in accordance with alignment information given on Drawings. Survey information issued by the S.O. shall be checked by the Contractor and discrepancies reported.

(ii) The Contractor shall be responsible for driving the tunnels and positioning linings and other structures within the specified tolerances.

(iii) All setting out and alignment control shall be carried out by suitably qualified and experienced persons. Specialist surveyor and laser guidance survey instruments will be required for the curve pipe-jacking.

(iv) Throughout the tunnel(s) any changes in gradient or alignment, including correction of errors, are to be effected by circular or transition curves, unless otherwise provided for in the design, or agreed with the S.O..

(v) The Contractor shall submit a method statement for setting out and controlling the alignment of the tunnels and shafts for agreement by the S.O. prior to commencement of work. Adequate opportunity for checking of the survey and setting out shall be given to the S.O..

(vi) Details of all setting out readings and calculations shall be maintained at the Site and be set out in a clear and orderly fashion to allow for ease of checking.

(vii) The Contractor shall provide all necessary lines and levels together with all necessary equipment to enable the Works to be constructed to the required tolerances. The setting out lines and levels shall be carefully preserved, and where necessary replaced, throughout the period for which they are required.

(viii) The primary setting out system shall be used for the main control of the excavation, tunnel lining and shafts, including checking and recording of the as built position of the lining at intervals.

G3.42.4 Groundwater

(a) Lowering of the natural groundwater table shall not be permitted.

(b) The Contractor's working methods and systems shall be designed to control groundwater to permit the construction of shafts, tunnels, breakouts and connections. The Contractor shall submit proposals to the S.O. for agreement prior to commencing installation and running of any dewatering system.

(c) Removal of groundwater shall not cause damage to the Works, or to third party property and shall not cause nuisance either by the removal of ground or by settlement.

(d) In planning temporary pumping systems, the Contractor shall take due consideration of water quality, pressure, quantity and variations in water levels.

(e) The Contractor shall obtain the necessary consents from the appropriate authorities to abstract and dispose of groundwater.

(f) In areas of contaminated land and other possible water filled voids, the Contractor shall take account of the potential hazards of inundation of the Works.

(g) Any Temporary Works for the control of water shall be removed and the ground reinstated when they are no longer required.

G3.42.5 Break-outs from Shafts and Tunnels

(a) For the junctions of tunnels and shafts the Contractor shall submit, for the S.O.'s agreement a method statement describing the details of the work, with Drawings and calculations.

(b) Tunnel and shaft linings shall not be broken out until all necessary Temporary Works, including ground treatment, where appropriate, have been completed to ensure the structural and geotechnical stability of the Works.

(c) In segmentally lined shafts, segments shall not be removed nor lining broken open until the temporary supports and/or ground treatment processes have been properly installed.

(d) Temporary supports shall not be removed until so agreed by the S.O. and the work for which they are required has been completed and the structure is capable of carrying the imposed loads. Once junction work is commenced it shall be completed as expeditiously as practicable.

(e) The break-out shall be of sufficient dimensions to accommodate the permanent structure to be built. Working space shall be kept to the minimum dimensions practicable.

(f) The break-up for a tunnel shall enable the construction of a stable initial length of an adequate tunnel lining, built within the specified construction tolerances.

(g) Ground improvement or other acceptable means shall be provided at shaft break-outs and break-ins as required to prevent instability of the ground as the sewer pipe excavation exits or enters a shaft and elsewhere as required.

G3.42.6 Grouting of Pipe Annulus

(a) The Contractor shall be fully responsible for preventing the occurrence of voids outside the pipe and if they occur he shall fill them with cement grout or other materials approved by the S.O..

(b) Immediately following the jacking operation, the Contractor shall pressure grout the jacked section to fill all voids existing outside of the pipe. Grouting shall be from the interior of the pipe through grouting holes as specified.

(c) Systems of standard pipe, fittings, hose, and special grouting outlets embedded in the pipe walls shall be provided by the Contractor. Care shall be taken to ensure that all parts of the system are maintained free from dirt. Grout composed of cement, sand and other approved compound and water shall be forced under pressure into the grouting connections at the invert and shall proceed until grout begins to flow from upper connections. Connections shall then be made to these holes and the operation continued to completion.

(d) Apparatus for mixing and placing grout shall be of a type approved by the S.O. and shall be capable of mixing effectively and stirring the grout and then forcing it into the grout connections in a continuous uninterrupted flow.

(e) After grouting is completed, pressure shall be maintained by means of stop cocks, or other suitable devices until the grout has set sufficiently. After the grout is set, grout holes shall be completely filled with dense concrete and finished neatly without evidence of voids or projections.

(f) For pipes with HDPE linings, grouting shall be carried out and the Contractor shall ensure that the pipeline is watertight before proceeding with the jointing of the linings of the pipes and the patching of the lining over the filled grout holes.

G3.42.7 Grouting of Disused Pumping Mains

(a) General

(i) The Contractor shall determine the pipe length and volume to be grouted, review the profile of the pipeline, clear any blockage or obstruction within the pipeline and identify appropriate locations along the pipeline for the entry of the grout (i.e. entry pit) and the grout-overflow outlet (i.e. exit pit) for each section of the pipeline to be grouted. The entry and exit pits can be at existing chambers or cut-sections of pipes along the pipeline. The pumping of grout shall be carried out from the highest point of the pipeline as far as possible.

(ii) The work zone shall be demarcated and isolated for safety.

(iii) Trial trenches shall be carried out when necessary to confirm the exact location of the pipeline.

(b) Grouting Process

(i) The Contractor shall comply with the relevant legislation, in particular the Workplace Safety & Health (Confined Spaces) Regulations which includes but is not limited to the following:

- Implementation of confined space entry permit;
- Evaluation of confined space for safe entry;

- Monitoring to ensure safety and health of worker during entry into or work in confined space;
- Periodic testing of atmosphere in chamber/manhole for unsafe atmospheric conditions;
- Adequate lighting in confined space;
- Ventilation of confined space;
- Appointment of confined space safety assessor;
- Conducting periodic emergency evacuation and rescue exercise;
- For works to be carried out on or affecting road carriageways, the Contractor shall comply with LTA's "Code of Practice for Works on Public Streets" and "Code of Practice for Traffic Control at Work Zone";
- A minimum 2 blowers, one for forced induction and the other for extraction shall be provided for workers working in the confined space. This is to ensure that constant fresh air is circulated in the confined space;
- The Contractor shall ensure that the works carried out shall not create any sparks that may cause possible fire hazard or explosion;
- The Contractor shall conduct Risk Assessments to identify all hazards, evaluate risks and plan for mitigating measures;
- The Contractor shall provide portable gas/vapour monitors for continuous monitoring of the atmospheric conditions inside the confined space;
- The Contractor shall provide air-release/grout-overflow outlets at the apex of the disused pipe for the release of built-up air or gases within the pipeline and for monitoring the effectiveness of the grouting works;
- The Contractor shall ensure that the grouting operation is continuous and the supply of the grout material is uninterrupted to prevent setting of the grout material inside the disused pipe prior to the completion of the grouting operation. This is to prevent voids in the disused pipeline. The grouting operation shall continue until the disused pipe is completely filled;
- The Contractor shall carefully plan the amount of grout needed for each operation and plan the delivery schedule with his supplier to avoid working beyond the approved hours; and
- The grouting operation shall be stopped when the grout material overflows from the air-release/grout-overflow outlets provided at exit pits and the grout material cannot be pumped any further into the disused pipe.

(c) Use of Existing Chambers as Entry/Exit Pits

(i) The Contractor shall note that when using an existing chamber as entry or exit pit, the Contractor is not permitted to dislodge the blank flange cover or fitting of the pipeline from its original position when there are persons inside the chamber. The Contractor shall prepare method statements, safe work procedures and risks assessments for the entire grouting procedure and submit them to the S.O. for approval.

(ii) Generally, the displacement of the cover/fitting shall only be carried out by persons outside the chamber using lifting belts attached to the cover/fitting. The Contractor shall ensure that all the bolts and nuts of the cover/fitting in the chamber are loosened and removed manually using hand tools only. If there is a need to use filing or manual sawing tool, the Contractor shall implement measures to prevent the generation of sparks/heat. After attaching the lifting belt to the cover/fitting, all persons shall leave the chamber before the displacement work is allowed to be carried out.

(iii) To prevent the generation of sparks/heat during the displacement process, the Contractor shall ensure that the chamber is filled sufficiently with water to submerge the cover/fitting before displacing the cover/fitting.

(iv) After displacing the cover/fitting, the Contractor shall ensure that the chamber is again evaluated for safe entry of confined space and if found to be safe, the remaining water in the chamber can be pumped out and persons can re-enter the chamber and continue preparation works for grouting. The procedure and entry permit system for working in confined space shall be maintained throughout the duration of the grouting process.

(d) Use of Cut-Sections of the Pipeline as Entry/Exit Pits

(i) The Contractor shall ensure that approvals from the relevant authorities and the S.O. are obtained for the excavation works to expose the section of the pipeline to be cut for use as an Entry/Exit pit.

(ii) Before cutting any section of the disused pipeline, the Contractor shall remove the blank flange cover or fitting of the pipeline at the existing chambers upstream and downstream of the Entry/Exit pit and ensure that the disused pipeline is properly ventilated and certified free of flammable gases. The procedures on the removal of the cover/fitting at the chamber are as described in the above Sub-Clause for "Use of Existing Chambers as Entry/Exit Pits". As an added precaution, the cutting of the disused pipeline shall only be carried out using "Cold Cutting" method. Flame cutting, sawing and other mechanical means which might produce heat/sparks are strictly prohibited because of potential fire hazard or explosion.

G3.42.8 Shafts

(a) Shafts - General

(i) The excavations for manholes/chambers may be used as working shafts. Where a working shaft is used for the construction of a manhole/chamber, the Contractor shall allow for this in his rates for construction of the manhole/chamber. Additional working shafts may be sunk with the approval of the S.O. but will not be paid for. Excavation shall comply with the Clause "Excavation for Tunnels" of this Specification.

(ii) The Contractor shall submit for approval to the S.O. dimensioned Drawings and calculations of the timbering or trench sheeting details for the working shafts. Each working shaft shall have a separate ladder bay for access which shall be isolated from the part of the shaft used for hoisting materials.

(iii) Prior to construction of shaft and excavation, the shaft area shall be thoroughly checked for existing utilities and the appropriate measures taken in agreement with the S.O.. All temporary support and shifting of existing utilities and services shall be deemed to be included in the rates.

(b) Shafts – Safety

(i) Shafts shall be provided with guard rails or rings at least 1.2 m above the surrounding ground level. All shafts shall be protected against the risk of flooding during construction by the provision of bund walls at ground level around by the top of the shaft or other measures as agreed with the S.O..

(ii) The shafts shall be equipped with ladders and landings at intervals not exceeding 6m. All ladders in the shaft are to be properly secured.

(iii) Should heavy plant and heavy loads have to be located in close proximity to any shaft, the Contractor shall take into account the effects of these loads as well as any ground loads when designing the shaft.

(iv) The Contractor shall guard against distortion of shaft lining during construction and the possibility of shaft settlement or uplift at an intermediate stage of construction.

(c) Temporary Shafts

(i) Full details of temporary shafts required by the Contractor's shafts working methods shall be submitted to the S.O. for his agreement. Such shafts shall be adequately sized for all operations required for the execution of the Works. Approval by the S.O. shall not relieve the Contractor of his sole responsibility for the efficacy, reliability and soundness of the method employed in completing the Temporary Works in a satisfactory manner.

(ii) The Contractor's Professional Engineer (Geotechnical, if applicable) shall design and endorse the Temporary Works design calculations and Drawings. These have to be checked by the Accredited Checker (Geotechnical) separately engaged by the Board for review and agreement in accordance to the Building Control (Amendment) Act 2007. The Contractor shall also submit the Temporary Works design calculations and Drawings to the S.O. for review and concurrence before submitting to the Authority.

(iii) During the construction of the temporary shafts, the Contractor shall seek his P.E. (Geotechnical)'s approval at every stage of the excavation, including the removal of struts during backfilling and seek the S.O.'s agreement before proceeding to the next stage of work. All statutory forms relating to the earth retaining stabilising structures shall be duly endorsed and submitted to the S.O. and Authority within the stipulated time given by the Authority.

(iv) Backfill for temporary working shafts shall comprise compacted materials complying with the Clause "Earth filling" of the Section "Specification for Materials" or as approved by the S.O..

(v) Where the Contractor wishes to recover temporary shaft linings the structure shall be removed in safe stages as backfilling proceeds, taking care to maintain the safety and structural integrity of the remaining lining. No part of Temporary Works shall be left in the ground within 2 m of the designed final surface level or otherwise specified by the property owners or Agencies. The Contractor shall be deemed to be included in his rates for completely removal of the temporary shaft whenever instructed by S.O. or respective authority & agencies.

(d) Shafts – Construction

(i) Shaft sinking shall be carried out by a method suitable for all the particular circumstances of the Site including ground parameters, groundwater, depth and final purpose.

(ii) Where work is done by underpinning, only that ground which may be safely excavated for the installation of one ring or one unit of support shall be carried out. Until that depth is properly secured by permanent or temporary shaft lining, no further excavation shall take place. In the case of pre-formed rings securing shall include grouting.

(iii) Where work is done by a caisson operation, a cutting edge shall be fitted to the leading ring. The cutting edge shall be maintained at an even level all round during shaft sinking. Jacking arrangements or kentledge shall be adequate for the work. The Contractor's arrangements shall ensure the stability of any kentledge. A lubrication space shall be maintained around the full shaft periphery during sinking.

(iv) Concrete walls installed by slurry trench or secant piling techniques shall comply with best practice. The Contractor's proposals shall be submitted to the S.O. for his agreement.

(v) Where low pressure compressed air is used to assist shaft construction, air decks and locks shall be designed by the Contractor to give adequate margins of safety against the air pressures to be used in the shaft. The Contractor's proposals shall be submitted to the S.O. for his agreement.

(vi) Full details of Temporary Works required by the Contractor's method of working for the construction of tunnel or pipe-jack eyes in the shaft lining shall be submitted to the S.O. for his agreement.

(vii) Shaft bases shall be concreted as shown on the Drawings. In the case of Temporary Works shafts the Contractor shall submit his proposals for the shaft base structure taking account of ground and groundwater forces and sealing the shaft against water entry.

(viii) In the event that dewatering or grouting for ground treatment is employed the work shall comply with the Section “Specification for Soil Stabilisation” of the Specifications. Water leakage through shaft wall or joints shall be plugged before proceeding to the next stage of shaft construction.

G3.42.9 Steel Trench Sheeting (Sheet Piling)

(a) Wherever the sub-soil conditions are expected to be of a soft and unstable character, the normal methods of timbering will probably prove insufficient to avoid subsidence of the adjoining road surfaces, concrete drains and canal and other services.

(b) In such circumstances, the Contractor will be required to use steel trench sheeting or steel sheet piling adequately supported by steel struts, walers, preloading, etc.

(c) Steel sheet piles shall conform to the provision of BS 7668:” Weldable structural steels. Hot finished structural hollow sections in weather resistant steels”, BS EN 10025-1: “Hot rolled products of structural steels”, BS EN 10029 “Hot-rolled steel plates 3 mm thick or above” Grade S275 or approved equivalent grade and shall be driven where required and approved or directed by the S.O..

(d) The Contractor will be required to design the layout of the piling and the overall dimensions of the excavation to suit the sheets and corners available and to give sufficient working space for the proper construction of the work.

(e) The Contractor will be expected to supply, pitch, drive and subsequently remove trench sheeting or piling in accordance with other items of the Specification and the terms "timber" or “timbering" shall also apply to steel trench sheeting or steel sheet piling throughout. Voids form arising from removal of sheet-piles shall be filled with sand or grout to minimise settlement.

(f) All temporary trench support must be designed by the Professional Engineer (Geotechnical, if applicable) engaged by the Contractor. The Professional Engineer (Geotechnical, if applicable) must visit the Site on a regular basis to ensure and certify the adequacy of the construction of the Works designed by him.

G3.42.10 Lighting and Electrical Works

(a) When work is in progress, the excavated tunnel shall be lit by flame proof lamps operating at not more than 55 volts and fed from a double winding transformer with the centre-tap of the secondary winding earthed. Lamps shall be provided at not more than 5 m intervals and naked lights shall not be used. All electrical installations shall be protected by a current ELCB and shall comply with Singapore Standard SS CP:5. The Contractor shall engage a licensed electrical worker to oversee all electrical work.

(b) Floodlighting on the Site surface shall be adequate for the safe operation of the Site. It shall be sited where necessary to ensure the light is directed to areas within the Site, and to avoid nuisance.

(c) Lighting in the tunnel shall extend the full length and not be less than that required for safe working and access.

(d) An alternative source of power and an emergency lighting system shall be provided to allow emergency securing operations and evacuation safely in the event of a primary power failure. An adequate number of hand lamps shall be located and maintained to working order at all times, at key points underground.

G3.42.11 Verifying Invert Levels of Existing Sewer Pipes

(a) The proposed sewer pipes shall be laid as far as practicable to lines and levels as indicated in the Drawings.

(b) To ensure that the new sewer pipes will be able to connect level-wise to the existing sewer pipes, the Contractor shall, prior to the commencement of excavation work, set out the proposed lines and verify the invert levels of existing sewer pipes.

(c) All levels taken shall be established with Singapore Land Authority (SLA), Precise Levelling Datum (PLD) or Precise Level Bench Mark (PLBM), which is reckoned at 0 m SHD as datum. Adequate temporary bench marks shall be transferred from the above to the work Site. In general, the invert level of an existing downstream manhole shall be adopted as the Master Bench Mark for the Contract. Where the invert of the existing downstream manhole based on above differs from the Board's record, the Registered Surveyor shall notify the S.O. who shall decide on the value of the bench mark to be adopted. The adopted bench mark shall then be the Master Bench Mark for the Contract.

G3.42.12 Connection to Existing Manholes

(a) The Contractor shall note that in connecting pipelines to existing manholes, every care shall be taken to ensure that the connections are watertight and the existing sewer pipe and manholes are not damaged. Special precautionary measure shall be taken to ensure the stability of the existing manholes, vortex drop chamber and sewer pipes connected to them.

(b) In connecting the pipeline to the existing manhole, a circular hole large enough to accommodate the pipeline shall be made in the wall and the cut end of the pipe neatly rendered to form a smooth bore.

(c) The Contractor shall ensure that there is no interruption to existing flows during the connection Works. No debris shall be allowed to fall or be discharged into the existing sewer pipes. Any debris which falls into the existing sewer pipes and any obstructions thereby caused shall be removed at the Contractor's expense.

(d) The Contractor shall not allow used water to flow into the newly constructed pipeline or seal off the existing pipelines until and unless the part of the pipeline concerned and the downstream pipelines have been completed, inspected and certified fit for use. Before doing so, the Contractor shall wait for direction from the S.O. in writing.

(e) The Contractor shall allow for all the above in his rates.

G3.42.13 Connection to Existing Pipes

(a) Where connections are to be made to existing pipes, the Contractor is to ensure a compatible fit between the new and the existing pipes.

(b) Where such a fit is not possible, the Contractor shall be required to obtain the S.O.'s approval for the use of a specially cast/manufactured piece for the transition. The cost of such special pieces is deemed to have been included in the Contractor's priced rates.

G3.42.14 Temporary Diversion of Existing Sewers

(a) The Contractor shall note that some manholes and sewer pipes are to be constructed onto existing sewer/pumping mains. He shall take all precautionary measures to ensure no interruption to the flow in the existing sewer. Care shall also be taken to prevent any debris falling into the existing sewer and being carried downstream. Any debris that has fallen into the existing sewer must be removed immediately at the Contractor's expense. The Contractor shall not seal off the existing sewer until the new temporary sewer diversion line has been completely inspected and certified fit for use. For such cases, the Contractor shall plug up the ends of the newly constructed temporary sewer pipes until diversion of flow can be implemented. The breaking up of the existing sewer, blocking of flow etc., shall be done during night hours.

(b) The Contractor shall ensure that no drain lines affected by the work are sealed off without prior and proper diversion of the affected drain lines to the satisfaction of the S.O.. Diversion and/or sealing off Works shall not be carried out until notice has been given by the S.O.. The Contractor shall allow for all the above in his rates.

G3.42.15 Working in Compressed Air

(a) The usage of compressed air facilities shall, in addition to complying fully with all the requirements of the Workplace Safety and Health Act, the Factories (Medical Examinations) Regulations and the Factories (Building Operations and Works of Engineering Construction) (Amendment) Regulations 2004, comply with these requirements of Specifications.

(b) Where the Contractor's proposals incorporate the use of compressed air, he shall ensure that the compressed air plant is available before work is allowed to commence.

(c) The compressed air plant shall consist of a main and a standby plant. The main plant shall be electrically driven from a Power Grid mains supply, whilst the standby plant shall be operated by other means such as on-site generator. The standby plant shall be used in lieu of the main plant to supply the tunnel with air for a minimum period of 1 hour each week.

(d) The S.O. may at any time during the use of compressed air, require tests to be carried out to determine the minimum pressure needed. For this purpose, the air pressure shall be reduced or increased in stages as ordered by the S.O.. During the test period, observations will be made on the conditions of the working face. The Contractor shall not be entitled to any extra payment for carrying out such tests.

(e) When the S.O. agrees that the work can be continued in free air, he may order that the compressed air plant be kept available at no extra cost.

G3.42.16 Specification for Working in Compressed Air

(a) This specification shall be read in conjunction with the specifications on Pipe-jacking. Compressed air working, compressed air plant, air locks and all other installations shall comply with the requirements of BS 6164 (Code of Practice for Safety in Tunnelling) and the UK Health and Safety Executive Work in Compressed Air Regulations with accompanying Guidance Notes.

(b) It may be necessary for the Contractor to work in compressed air in conjunction with heading, tunnelling or pipe-jacking if the ground conditions necessitate it. The Contractor is required to engage a Professional Engineer (Geotechnical, if applicable) to submit all his proposals for the use of compressed air in heading, tunnelling or pipe-jacking and to obtain the approval of the S.O. before work starts on this section.

(c) The Contractor shall submit to the S.O. complete details of the following:

(i) Bulkhead, air lock, man-lock, medical lock if the working pressure exceeds 1 bar, and other structures used in connection with work in compressed air;

(ii) Plant and equipment for the production and supply of compressed air and standby arrangements complete with pipe-work; including dehumidifier and chiller etc. air flow control valves; pressure gauges; thermostatic control system and thermometers;

(iii) Emergency exit from and entrance to working chambers; alarm system; fire-fighting equipment;

(iv) Electrical control panel, lighting system and DC emergency reserve circuit*;

(v) Comprehensive communication system;

(vi) Schedule of working pressures; compression and decompression procedures;

(vii) Schedule of work shifts;

(viii) Ventilation system including oxygen and explosive gas meters;

(ix) Noise suppression measures for the plant and equipment as well as for the air-lock, man-lock;

(x) Name and address of medical officer or hospital to which workers will be referred;

(xi) Location of nearest decompression chamber (medical lock);

(xii) Any other details as directed by the S.O.; and

(xiii) * Note:

1) All electrical power supply shall be at 100 volts, 50 Hz.

2) All lighting shall be sodium vapour type and explosion proof.

(d) The Contractor shall submit full details of the operational procedures that will be followed in working in compressed air and to obtain the approval of the S.O. before work starts on this section.

(e) The Contractor shall, after installation and prior to commencing work in compressed air, subject the plant to a 24-hour continuous running test during which it must operate to the satisfaction of the S.O.. All plant and equipment shall be maintained in satisfactory working conditions at all times.

(f) The Contractor shall not employ any person on work in compressed air unless he has been certified fit for such employment by a medical officer not more than 3 days before commencing work. No person suffering from a cold in the head, chest infection, sore throat or earache shall work in compressed air and any person suffering from these conditions or any other illness or injury necessitating absence from work for more than 3 days must be re-examined and certified to be fit by a medical officer before he resumes work in compressed air. During employment in compressed air, a person must be re-examined by a medical officer once every 3 months if the maximum working pressure does not exceed 1 bar, or, once every 4 weeks if the maximum working pressure exceeds, or is liable to exceed, 1 bar. The Contractor shall be responsible for arranging these medical examinations at his own expense.

(g) The Contractor shall not employ any person on work in compressed air unless the person has had previous experience in such work, or, if he has not had such experience, is under the supervision of a person experienced in such work. In the case of a person not previously employed in compressed air, compression shall not be carried out unless he is accompanied in the man-lock by a person competent to advise him on the appropriate conduct of persons during compression. When the working pressure is 1 bar or above, new workers shall not work more than 4 hours on the first working day.

(h) The Contractor shall fully inform all compressed air workers of the hazards of working in compressed air. The Contractor shall supply all compressed air workers with pamphlets containing advice as to precautions to be taken and regulations to be obeyed in connection with such work. If the working pressure exceeds 1 bar, the Contractor shall supply all the workers with suitable durable labels, to be worn next to the body at all times, indicating that the wearer is a compressed air worker and giving the address of the medical lock to which he shall be conveyed if taken ill. In particular, the Contractor shall inform all compressed air workers that consumption of alcohol and smoking are strictly forbidden in all compressed air areas at all times.

(i) The Contractor shall make arrangements to ensure that whilst any person is in compressed air, a person competent to take charge of compressed air operations is in charge of the operations and in attendance on the site.

(j) The Contractor must ensure that, when the working pressure is 1 bar or above, each person must spend at least 12 consecutive hours at atmospheric pressure and not be subjected to more than 3 exposures to pressure in any 24-hour period.

(k) Every man-lock shall be in the charge of a competent lock attendant when any person is in the man-lock or in a working chamber to which the man-lock affords direct or indirect access. The lock attendants must be familiar with the procedures and problems associated with compression, decompression, decompression sickness and the keeping of records.

(l) When the pressure in a working chamber exceeds 1 bar, the Contractor shall provide and maintain a suitably constructed medical lock situated as near as possible to the man-lock. The medical lock shall at all times be kept ready for immediate use. A medical lock attendant must be on duty at all times when the working pressure is 1 bar or above and for 24 hours after the last man-lock decompression for pressures of 1 bar or above and when any patient is in the medical lock. The medical lock attendant shall be a qualified nurse, hold a current certificate of proficiency in first aid and be trained in the medical aspects of compressed air work.

(m) The Contractor shall provide and maintain for the use of all compressed air workers the following:

(i) Clothes lockers and adequate and suitable facilities for the changing and drying of clothes and footwear;

(ii) Adequate and suitable facilities for washing, including soap and clean towels and showers with hot water;

(iii) Adequate and air-conditioned facilities for remaining on Site for 1 hour after decompression from pressures exceeding 1 bar; and

(iv) Chemical closets in all working chambers, medical locks and man-locks where decompression time exceeds 30 minutes;

(n) The Contractor shall maintain for all compressed air workers a proper record of the following:

(i) Individual results of all medical examinations;

(ii) Accurate details of all cases of decompression sickness; and

(iii) Details of each exposure to compressed air of each person.

(o) The Contractor shall submit relevant extracts of the records to the S.O. as and when directed. The Contractor shall indemnify the Board / S.O. against all claims that may be made by the workers for working in compressed air.

(p) The Contractor shall allow in his Tender price for work in compressed air for complying with all the above requirements and the latest requirement by the Industrial Health Division of the Ministry of Manpower on the use of compressed air and all other requirements which he deems necessary to complete the work successfully and no claim to the contrary will be entertained by the Board.

G3.42.17 Tolerances for Tunnels, Shafts and Underground Works

(a) Tolerances - All Tunnels and Shafts

(i) Tunnels and shafts shall be constructed to the centre lines and shafts required by the Contract and subsequently agreed on Site with the S.O..

(ii) Overall tolerances allowed in respect of the departure of any point on the internal profile of the structure from its established centre line (s) shall be as given in the Table below. The tolerance includes all building errors.

	Tunnel Dia. (m) <5m
Pipe-jacking: Line	±50mm
Level	±25mm

(iii) Where there are hydraulic requirements, no backfalls forming localised ponding shall be allowed.

(iv) Shafts shall be sunk to a maximum divergence of 1:300 from the design centre line at any depth.

(b) Tolerances - Segmental Shafts

(i) The maximum lipping between the edges of adjacent segments shall be 5 mm.

(ii) The plane of each segmental ring shall not depart at any point from the plane surface normal to the longitudinal axis by more than 10 mm.

(iii) Where the diameter is greater than 2 m, the maximum and minimum measured diameters in anyone ring shall be within 1% of the theoretical design diameter of the ring.

G3.42.18 Surveying and Setting Out

(a) Surveying and Setting Out - Datum for the Works

(i) The level datum for the Works shall normally be the agreed Works national datum or as stated in the Contract.

(ii) The plan survey shall be conducted to a Works grid by a registered license surveyor established over the area of the Works. Unless otherwise given in the Contract, the orientation of the Works grid shall relate to the national survey grid.

(b) Surveying and Setting Out - Survey Bench Marks

(i) The Contractor shall install all level and survey stations marks required. Such stations shall be of robust construction, protected against damage and the influence of any movement which may arise from the execution of the Works.

(ii) The Contractor shall check the condition and resurvey survey stations at intervals during the progress of the Works.

(iii) The Contractor shall provide the S.O. with the location and description of all survey stations, the results of surveys and all calculations. Where required, he shall give adequate opportunity for the S.O. to check such stations prior to their utilisation.

(iv) The degree of accuracy employed in the survey and setting out shall be such as will allow the alignment, levels and dimensions specified for the Works to be achieved.

(v) The Contractor shall ensure that all surveying equipment used for the Works is properly maintained and that the performance of the equipment complies with the manufacturer's specification for accuracy. The Contractor shall provide a certificate to confirm the maximum allowable jacking load. The Contractor shall be responsible for calculating the maximum allowable in-situ jacking loads based on the soil information provided in the tender documents and shall also ensure that these loads are not exceeded.

G3.42.19 General Requirements for Demolition Works

(a) This Specification is to be read in conjunction with the Drawings, Specifications and Bills of Quantities and all other documents in respect of this tender.

(b) The works shall be carried out in accordance with the SS557:2010 – Code of Practice for Demolition, SS CP 14:1996 – Code of Practice for Scaffold, SS CP 49:1998 – Code of Practice for Noise Control on Demolition Sites, BS 7121 – Code of Practice for Safe use of Cranes and any other relevant Singapore or British Standards current on the date of invitation to tender. Any reference in the Specification to a Standard published by the British Standards Institution, or to the specification of another body, shall be construed equally as reference to an equivalent one.

(c) Any clauses of this Specification which relate to work or materials not required in the Contract shall be deemed not to apply.

(d) The Contractor shall present his programme of operations in his tender submission and submit to the S.O. for approval before starting to work. The S.O. may vary the programme or adjust the quantity of the work during the course of the operation at his discretion and the Contractor shall adhere and execute the work in a manner satisfactory to the S.O..

G3.42.20 Demolition

(a) Method of Demolition

(i) The Contractor shall submit the method statements, demolition works plans and calculations to the Superintending Officer (S.O.) before commencement of Works. The Contractor shall comply with the procedures and recommendations as detailed in Singapore Standard SS557:2010 – Demolition, SS CP 14: 1996 – Scaffold, SS CP 49: 1998 – Noise Control on Construction and Demolition Sites, BS 7121 – Code of Practice for Safe Use of Cranes.

(ii) The method statements, demolition works plans and calculations are required for plan submission and application for the relevant permit to commence demolition works. The demolition works shall only commence upon receiving approvals from BCA and other relevant authorities.

(iii) The method statements, demolition works plans and calculations shall include but not limited to the following:

1) The demolition plans showing the detailed sequence for the demolition of aboveground and underground structures, movement of powered mechanical equipment, the amount of debris allowed on site, clearance path, means of transport and temporary propping / supports required;

2) The methods of testing and removal hazardous materials if any;

3) The methods of debris handling;

4) Special safety considerations; and

5) The equipment to be used for the demolition works; and

6) Calculations demonstrating the ability of existing structures to take temporary loads such as construction equipment loading, demolition debris loading etc.

(b) Standard Scope of Work for Demolition

(i) Removal and disposal of ALL aboveground structures such as buildings, tanks, panels, lifting frames, driveway, footpath, fencing etc.

(ii) Removal and disposal of ALL underground structures such as chambers, sumps, manholes, pilecaps, footings, pumping mains, sewers, drains, and other services, etc.

(iii) Hacking down of all existing piles by another 1 metre from the base of existing pilecaps.

(iv) The presence of asbestos material shall be assessed by a Competent Person for Asbestos Survey. If present, asbestos material shall be removed by MOM Approved Asbestos-Removal Contractor (AARC) and disposed by NEA approved asbestos disposal contractor.

(v) Removal and disposal of remaining sludge in the chambers/manholes/tanks and all debris resulting from the demolition works to approved dumping grounds.

(vi) Tankering and disposal of remnant used water within the pumping stations to water reclamation plants. The Contractor shall ensure that this requirement is strictly complied with. Under no circumstances shall remnant used water be discharged into the drain networks.

(vii) Removal and disposal of all trees and shrubs within the boundaries of the installation.

(viii) Plugging / sealing off of all abandoned services at sites' boundary.

(ix) Protection of all trees, services, chambers, sewers, manholes, fencing and gate that are to be retained if any, during the course of demolition works.

(x) Upon completion of the demolition works, all sites shall be backfilled with approved backfill materials, compacted, levelled with a fall to drain off water, and close turfed with cow grass.

(xi) Reinstatement and repair of damages to existing services, structures and turf, etc. as required for the complete execution of the Works.

(xii) Making good of existing drains, driveway, footpath, fencing, etc. to be retained if any, including replacement if damaged.

(xiii) Where specified in the Bills of Quantities, provision of Environmental Baseline Study (EBS). The Contractor shall comply to all technical guidelines and requirements as indicated in JTC website, http://www.jtc.gov.sg.

(xiv) Provision of certified cadastral survey plan of the sites after the completion of the Works by Contractor's Registered Surveyor. The position, dimension and level of all retained services/structures/piles within the boundary shall also be shown in the plan.

(c) Boundary Fencing

(i) In the event that the position of the existing fencing is not in line with the lot boundary shown in the Certified Cadastral Survey plan, the Contractor shall, when instructed by the S.O., relocate the fence to the exact lot boundary shown in the survey plan.

(d) Site Drainage

(i) The Contractor shall note that there must not be any water ponding on the completed turfed surface at anytime. The Contractor shall backfill the site in such a manner that a sufficient gradient (preferably 1:200) is provided for water to drain off naturally.

(ii) The Contractor shall submit the proposed backfill levels for the sites for S.O.'s approval prior to the commencement of backfilling works.

(e) Control of Dust Pollution

(i) Screen nettings with a minimum height of 2.4m shall be provided at the top of site hoarding to minimize dust pollution. At sensitive areas such as residential or commercial areas where there could be potential complains from neighbouring residents, the height of the netting above the hoarding shall be increased to 3.6m or more

(ii) Water shall be constantly sprayed during demolition works to prevent dust from getting airborne.

(f) Responsibility of Contractor's Professional Engineer

(i) The Contractor's P.E. shall be fully responsible for the design and supervision of all temporary works necessary to carry out the works, which may include but is not limited to the following:

(ii) Earth Retaining and Stabilising Structures.

(iii) Vehicular Traffic Deckings.

(iv) Working Decks.

(v) Temporary Access Slopes.

(vi) Temporary Supports.

(vii) Debris Protection Shelters.

(viii) Hoardings.

(ix) Dust Screens.

(x) The Contractor's P.E. shall design the temporary works in conformance to appropriate standards and codes of practice and latest requirements by statutory boards and Government Authorities. As a minimum, the Contractor's P.E. must consider the following aspects in his design and ensure the design is structurally safe:

1) Adequate site investigation;

2) Proper selection of the soil parameters for design;

3) Effects due to onerous water pressures and seepage forces;

4) Effects under both drained and undrained conditions of the soils, as well as the effect of time on soil drainage conditions;

5) Effects of surcharge loads, including incidental loads, construction loads, adjacent slope, adjacent structures, etc.;

6) Design robustness and redundancy considerations, that shall include one-strut failure, accidental loads, etc.;

7) Overall wall stability and basal heave stability;

8) Structural adequacy of supporting system, such as walers, struts, anchors, etc. including provision of adequate stiffeners;

9) Provisions of restraints in structural members' connections, ties and bracings;

10) Adequate wall embedment;

11) Varying load conditions during stages of the construction;

12) Effects of ground deformation on adjacent properties;

13) Effects due to ground water lowering; and

14) Sensitivity analyses and impact on the performance of temporary works.

(xi) The Contractor's P.E. shall allow for including but not limited to the following aspects in his design:

1) Surcharge load of at least 20kN/sq m, construction and incidental loadings, and loads from the adjacent existing structures and usage etc.;

2) Full water table level or onerous water conditions and seepage pressures acting on the wall;

3) Mobilization factors shall conform to SS EN 1997: "Eurocode 7: Geotechnical Design" requirements;

4) Unplanned excavation;

5) No allowance for any material overstress;

6) One-strut failure and accidental loads; and

7) Factor of safety for the design of temporary works shall not be less than that for permanent works.

(xii) The Contractor's P.E. shall carry out the followings:

1) Construction method and sequence.

2) Ensure that the construction plans, method and sequence, and the Contractor's detailed excavation plan are fully in accordance to the design for temporary works.

3) Instruct the Contractor to seek approval from the Contractor's PE before proceeding with any excavation at every strut level.

(xiii) Inspection of erected works

1) Check that an adequate and proper instrumentation plan is executed on site before the commencement of any excavation or temporary works.

2) Inspect and assess the performance of the wall at every strut level before giving approval to the builder for further excavation.

3) Carry out adequate site inspections, to ensure that all structural elements are adequately installed and constructed in accordance to the design which is verified by the actual site conditions.

4) Monitor closely the site works, inspect and ensure that all deformations, loads and pressures do not exceed allowable limits. The Contractor's P.E. shall take all the necessary preventive, protective and remedial actions on site in consultation with the the Board, the S.O. and the Contractor, to limit the wall and ground deformations.

(g) Drawings of Existing Installations

(i) Where available, drawings of existing installations such as pumping stations, pump sumps, solids divertor, sewage treatment plants, R2, holding tank etc. will be provided. The Contractor shall note that the layout and structure of the installations shown in the drawings provided are only indicative and shall be only used as a general guide. The provision of these drawings does not absolve the Contractor from the responsibility of carrying out detailed site investigations and making his own interpretation and judgement on the condition and scale of the structures to be demolished and the extent of temporary works and reinstatement works.

G3.42.21 Turfing

(a) General

(i) All turf disturbed or destroyed by excavation, site huts, dumps of materials, lorries and building work, etc., shall be carefully reinstated as soon as possible and watered and rolled, etc., until properly established and to the satisfaction of the S.O. / the agencies.

(b) Supply and Spreading of Topsoil

(i) All topsoil supplied by the Contractor shall be of approved quality, containing a large percentage of humus, free from rubbish, roots, stumps, stones or other extraneous materials, and well mixed.

(ii) The soil shall be laid with 50mm thick Axonopus compressus (cow grass) below the desired turf level with provision of 100mm depth planting mixture, The planting mixture should make up of 3 parts of loamy soil and 1 part of organic matter (processed woodchips or compost) and spread uniformly over the site; slopes and/ or berms etc. where directed and shall be lightly watered if necessary, and then tamped with wooden tampers and trimmed to the consolidated thickness required. The turf sods shall be planted using topsoil only.

(iii) The Contractor shall be responsible for providing and fixing the turf promptly after top soiling in order to prevent erosion thereof due to wet weather or other causes. Any loss of unprotected or inadequately protected topsoil shall be made good at the Contractor's expense. A thin layer of topsoil shall be broadcast on the turfs to fill all depressions and give a smooth finish.

(c) Supply and Fixing of Turfs

(i) For general turfing, Cow Grass (Axonopus compressus) shall be used.

(ii) Close turf shall be a live grass sod or mat at least 300 mm square with a well-developed root system growing in a minimum of 20 mm soil bed, free from stones or extraneous roots, cut mechanically by hand to give an extra thickness and texture.

(iii) All turfs supplied by the Contractor shall be of healthy and vigorous stock and from approved sources. All turves supplied shall be free from weeds, especially Mimosa pudica and care shall be taken to include as much of the root growth as possible.

(iv) Turf materials are to be obtained from a bona-fide horticultural source or private land owner. No turf is to be removed from any unauthorized locations, roadside, riverbank or private property without permission of the owner. The Contractor is to provide a Certificate to the S.O. to demonstrate the source of all turf delivered to the site before any turf is laid.

(d) Close Turf, Ground Preparation

(i) Rake the topsoil areas to a smooth and uniform grade free of any slight mounds or depressions to achieve a uniformly flat surface. Re-grade any depressions or humps that may occur until a satisfactory grade is achieved.

(ii) The area to be turf is to be brought to a fine tilt by approved mechanical means or by hand raking. Any stones over 25mm in diameter shall be removed from the site. Watering of the areas shall be carried out to produce a moist condition of the soil and to consolidate the soil.

(e) Close Turf, Planting Operations

(i) Close turf sods shall be laid onto the surface of the prepared ground with leaf turf upwards, but jointed as closely as possible to achieve a uniform cover. The turf shall be laid off planks working over turf previously laid. The whole area is then to be top-dressed with finely sifted topsoil mix to give an evenly smooth surface. The finished close turf shall be lightly compacted by treading or with a wooden beater to ensure even coverage and compaction.

(ii) Watering shall be undertaken by use of a fine spray to avoid disturbance of soil particles.

(iii) Close turf shall be only accepted as complete after the growth of an even sward is evident. Dead turf shall be immediately replanted with new turf.

(iv) Should there be any settlement due to lack of even compaction, this shall be corrected by application of top dressing by additions of new planting mixture. If the depression is greater than 25 mm, the depression and affected area shall be lifted, the depression filled with appropriate volume of planting mixture, lightly compacted and the affected area re-turf as specified. This operation shall be done as often as necessary to produce an even and smooth surface free from bumps and hollows.

(v) All close turf operation shall be carried out from wooden planks or plywood boards, with the workers moving away from completed turf areas, raking any compressed soil or footprints before lying of sods. All access onto soil areas shall be wooden boards or plywood sheets areas compacted by working are to be re-cultivated and re-laid.

(f) Maintenance of Turf

(i) The Contractor shall engage a qualified landscape supervisor to organize and manage the maintenance work. The Contractor's Supervisor shall inspect the site at least once fortnightly and shall prepare a brief checklist for the operations for the maintenance. A copy of the checklist is to be submitted to the S.O. fortnightly so that a running record of proposed operations can be checked during the maintenance inspections each month.

(ii) If in the opinion of the S.O. that the maintenance works have not been satisfactory carried out according to site conditions and the Specifications, part of the monthly payment will be withheld until the works have been satisfactorily carried out.

(iii) Throughout the maintenance period of the contract, all turfing shall be watered regularly and de-weed to the satisfaction of the S.O.. The contractor shall allow for such watering and tending. Any dead turves, badly growing turves or otherwise unsatisfactory turves shall be removed and replaced with approved quality turves by the Contractor at his own expenses. All weeds shall be removed at least twice a month.

(iv) About one month after planting, the grass shall be manure with a complete chemical fertiliser which supplies Nitrogen, Phosphorus, Potassium and magnesium approximately in the following proportion 18:12:6:3. This shall be applied at the rate of 125kg/ha of grass after all weeds have been removed. The fertiliser shall be mixed with topdressing and spread evenly over the field. This application shall be repeated after three months.

(v) After one month's growth, the grass shall be cut with a suitable rotary mower to encourage the grass to creep on the ground instead of growing upright. This operation shall be repeated at fortnightly intervals until the ground is completely covered with grass. A mechanised reel cut mower shall be used to cut the grass. This normally shall be done about three months after planting. Thereafter mowing shall be done fortnightly by reel cut mower until the end of the maintenance period.

(vi) Approximately two months after planting, when the grass has grown and produced a continuous cover over the whole area, the whole of the turfed areas shall be lightly rolled(roller shall not exceed 150kg in weight) to an even surfaces. Rolling shall be repeated at monthly intervals until the surface is accepted by the S.O..

(vii) Grass shall be closed cut as short as possible following the general contours of the ground. The stock or blade of the grass after each cutting shall not stand higher than 20mm above the ground level.

(viii) The edges of all side tables, and other grassed areas adjoining roadways, footpaths, kerbs, dividers and concrete paved areas shall be cut back to proper straight lines or curves as the case may be, and trimmed thereafter to produce a neat and tidy appearance. This work shall be carried out immediately after the grass in the area has been cut.

(ix) Grass growing in the cement joints of footpaths and along the edge of the roads adjacent to or directly fronting the sidetables, centre medians, etc. shall be cut or manually weeded out and cleared.

(x) All cut grass, cuttings together with trimmings, loose stones, rubbish etc. shall be collected and removed from the site and disposed of expeditiously.

(g) Final Handover Inspection

(i) Two weeks before the expiry date of the Maintenance Period, a joint inspection shall be held together with the Board and his Maintenance Agency and the S.O. to review the overall condition in order to obtain approval for Final Handover.

(ii) In order to ensure satisfactory handover procedures, the maintenance schedule submitted by the Contractor each will be reviewed to ensure adequate work has been done.

(iii) At the time of the final inspection, all areas under the contract shall be free of weeds, neatly cultivated and raked, and all plant are growing in an acceptable condition by the S.O.. Grass shall be neatly trimmed and all clippings removed. No bare patches of earth shall be visible in turf or planting areas.

(iv) If, after the inspection, the Board and the S.O. are of the opinion that all planting works and maintenance works has been performed in accordance with the drawings and specifications, the S.O. will give written acceptance and Handover of the project to the Board.

(v) If, certain portions of the work are not acceptable under the terms and intend of the Specifications, the formal maintenance period for all the work shall be extended at no cost to the Board until the defects in the work have been satisfactory corrected and rectified or accepted by the Board and S.O. at a later date.

END OF SECTION